

TOPIC 11. ADHESIVES

1. Introduction
2. Mechanisms of Adhesion
3. Classification. Types of adhesives
4. Design and processing considerations
5. Testing methods
6. Modes of failure
7. Degradation mechanisms
8. Examples

1. Introduction: Overview of joining methods

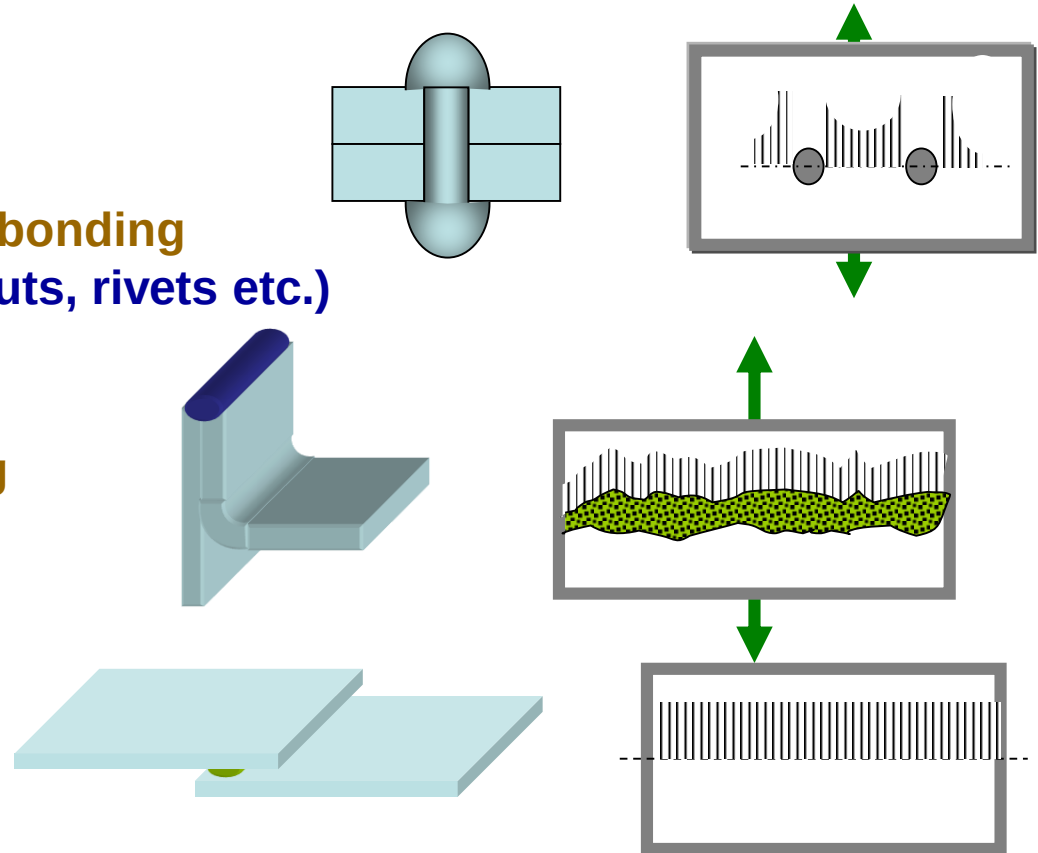
STRUCTURE: A group of **assembled** elements, behaving like a **whole** for mechanical stresses.

Joining methods:

- Physical systems: **Mechanical bonding** (fasteners, screws, bolts and nuts, rivets etc.)

- Metallurgical systems: **Welding**

- Chemical systems: **Adhesives**



Advantage of adhesives: uniform stress distribution

1. Introduction: Definitions

- **ADHESIVE:** is a compound or substance that adheres or bonds two items together.
- A substance that is capable of holding together other materials by surface attachment
- **ADHESION:** the property of sticking together or the joining of surfaces usually of different composition.
- State of two surfaces kept together through interfacial forces (intermolecular, chemical, mechanical or both)
- **ADHERENT:** the substrate over which the adhesive is applied

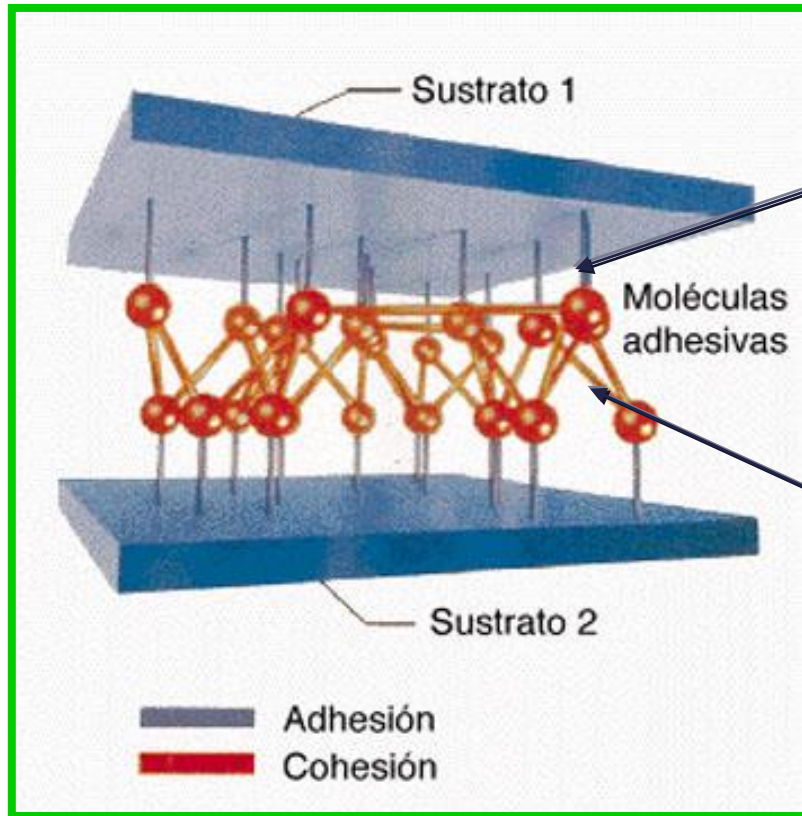
ADVANTAGES

- **Uniform stress distribution**
- **No distortion of substrates**
- **Bonding of dissimilar materials**
- **Simultaneous sealing**
- **Corrosion protection by isolation**
- **Reduction of the number of components**
- **Aesthetics: flatter surfaces, hiding joints**
- **Delicate and complicated components can be joined**

DISADVANTAGES

- **Surface preparation required**
- **Curing time**
- **Limited temperature capability**
- **Poor peel strength**
- **Limited reliability in hostile environmental conditions**
- **Toxicity**

2. Mechanisms of adhesion: Adhesion vs. cohesion



Adhesion: The adhesion mechanism between adhesive and substrate

Cohesion: the intermolecular force that holds together the molecules in adhesive

- Adhesive failure: failure that occurs between two phases
- Cohesive failure: failure that occurs within a phase

2. Mechanisms of adhesion

MECHANISMS OF ADHESION

- They try to explain the causes for adhesion between materials
- There is **NO** universal adhesion theory

Mechanisms of of Adhesion:

- Mechanical adhesion
- Electrostatical adhesion
- Diffusive adhesion
- Adsorption (physical adsorption)- dispersive adhesion
- Chemical adhesion (chemibSORption)

2. Mechanisms of adhesion

MECHANISMS OF ADHESION

- **Mechanical adhesion**
 - Oldest theory of adhesion
 - Based on adhesive interlocking mechanically in the surface roughness of the interface or pores of the substrate
 - Adhesive joints frequently possess an important mechanical component, but that it does not suffice as the sole mechanism whereby surfaces are joined
 - Favoured by:
 - Increase in substrate surface area due to increased roughness
 - Penetration of the adhesive in the pores of the substrate
 - Examples: Sewing (a large scale), Velcro
- **Electrostatical adhesion**
 - Applied to metal/adhesive joints where the different nature of the materials facilitates electron transfer from the metal to the adhesive to form a difference in electrical charge at the join
 - This results in a structure similar to a capacitor and creates an attractive electrostatic force between the materials.

2. Mechanisms of adhesion

MECHANISMS OF ADHESION

- Diffusive adhesion
 - Occurs when the molecules of both materials (adhesive/adherent) are mobile and soluble in each other
 - Explains adhesion between polymers by chain migration. Polymers must be soluble in each other
 - Is also the mechanism involved in sintering
- Adsorption (physical adsorption)- dispersive adhesion
 - Adhesion forces are due to interface molecular forces such as van der Waals forces or hydrogen bonding (secondary bonds) .
 - A van der Waals force is the attraction between two molecules that have positively and negatively charged ends.
 - Adhesion is produced when the adhesive wets the adherent. Good wetting is necessary

2. Mechanisms of adhesion

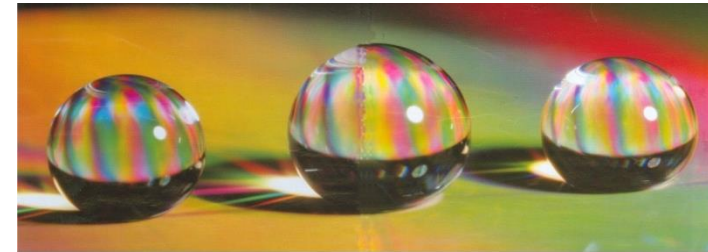
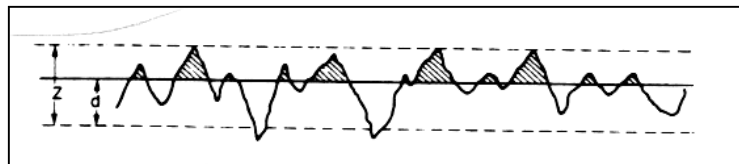
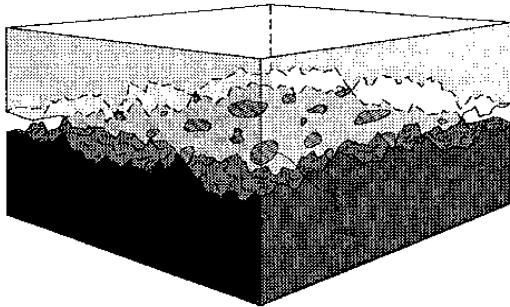
MECHANISMS OF ADHESION

- Chemical adhesion (chemisorption)
 - Creation of primary (ionic or covalent) bonds between the adhesive and adherent.
 - The strongest joins are where atoms of the two materials swap (ionic bonding) or share (covalent bonding) outer electrons. A weaker bond is formed if oxygen, nitrogen or fluorine atoms of the two materials share a hydrogen nucleus
 - Use of reactive adhesives such as epoxies or cyanoacrylics
 - Can be achieved through surface treatment of the substrate

2. Mechanisms of adhesion

Other factors that affect adhesion:

- Wetting –
 - Wetting of adherent by adhesive is required for good adhesion
- Surface roughness
 - increased surface roughness (by abrasion) improves bond strength.
 - Surface roughness:
 1. Raises contact area
 2. Removes surface contaminants
 - **However:** roughness must not be too high because air may remain trapped and the joint may lose its strength



MOJADO DEFICIENTE



MOJADO PARCIAL



MOJADO TOTAL

3. Classification. Types of adhesives

General components of adhesives

Base Product **Polymer**: Forms la mass of the adhesive

Additives

Substances that are added to modify the physical and chemical properties of the bulk polymer before and after the curing process :

1. Colorants
2. Antioxidants
3. Primers: low viscosity adhesive that is applied before making the adhesive joint in order to improve adherence. Reduces surface energy of adherent
4. Plasticizers : solvent for polymers. They separate polymer chains and enable deformation. Modify Tg
5. Tackifiers: improve tack(ability to stick) by reducing viscosity or improving wetting
6. Coupling agents: chemically reactive agent used to promote adhesion by absorption (chemical or physical)
7. Fluorescents substances.

Fillers

Substances used to modify the rheology of the adhesive.
Inert fillers to make final product cheaper or modify its density

3. Classification. Types of adhesives

According to the application:

- Structural:** must withstand and transmit high stresses from one surface to another
- Support, light-duty, fixing:** the maintain the weight of the adherent
- Sealants:** gap-filling adhesives

According to their temperature stability :

- Thermoplastics
- Thermosets

According to their composition:

- Animal origin: animal proteins, egg white, spider web, casein (milk protein) “glues”
- Vegetal origin: (“resins”) starch, dextrin (gummy substance obtained from starch), rubber, vegetable proteins
- Synthetic adhesives: derived from the chemical industry

According to their curing properties:

- Hot melts: (thermoplastics) solidification of a molten polymer
- Prepolymerized: solvent is evaporated
- High resistance or polymerizing: Adhesives that polymerize at the adhesive joint

3. Classification. Types of adhesives

According to their curing properties:

PREPOLYMERIZED: Solvent is evaporated . They contain a polymer in solution or suspension in a solvent. Paper glue, sticky tapes...

- When solvent evaporates $\Rightarrow$ decrease in volume
- It is difficult to remove all the solvent $\Rightarrow$ adhesion cannot be easily controlled

POLYMERIZATION occurs at the adhesive joint: Monomers that polymerize at the adhesion $\Rightarrow$ joining process can be controlled: **high resistance ADHESIVES**

RIGID

- **Anaerobic**
- **Cyanocrilycs:** Bad for vibrations.
 $T_{\max}$ 80°C.
- **Epoxies:** Very hard, bond almost any substrate, high strength, $T_{\max}$ 180°C.

TOUGH

- **Acrylics:** Tougher than epoxies

FLEXIBLE

- **Silicones:** need (room) moisture to cure.
- **Polyurethanes:** For car windscreens

If it's a polymer you can probably make an adhesive out of it

3. Classification. Types of adhesives

Anaerobics Adhesives

- **Single component** systems based on **acrylic resins** (thermosetting)
- Set in the **absence of oxygen** (usually in the presence of a metal)
 - Can set in the absence of temperature. Inhibited by oxygen
- *Good sealant and damping properties*
- *Brittle*
- *Limited to $T < 150^{\circ}\text{C}$*
- Example: Loctite® used for threaded fasteners

Cyanolacrylics

- **Single component** systems based on **acrylic resins** (thermosetting)
- React with **moisture** on surface
 - Harden in a few seconds in the presence of surface water
- fast with good adherence ✓
- High strength ✓
- rapidly evolving ✓
- brittle ✗
- poor moisture resistance ✗
- expensive ✗
- Example: “Super glue”

3. Classification. Types of adhesives

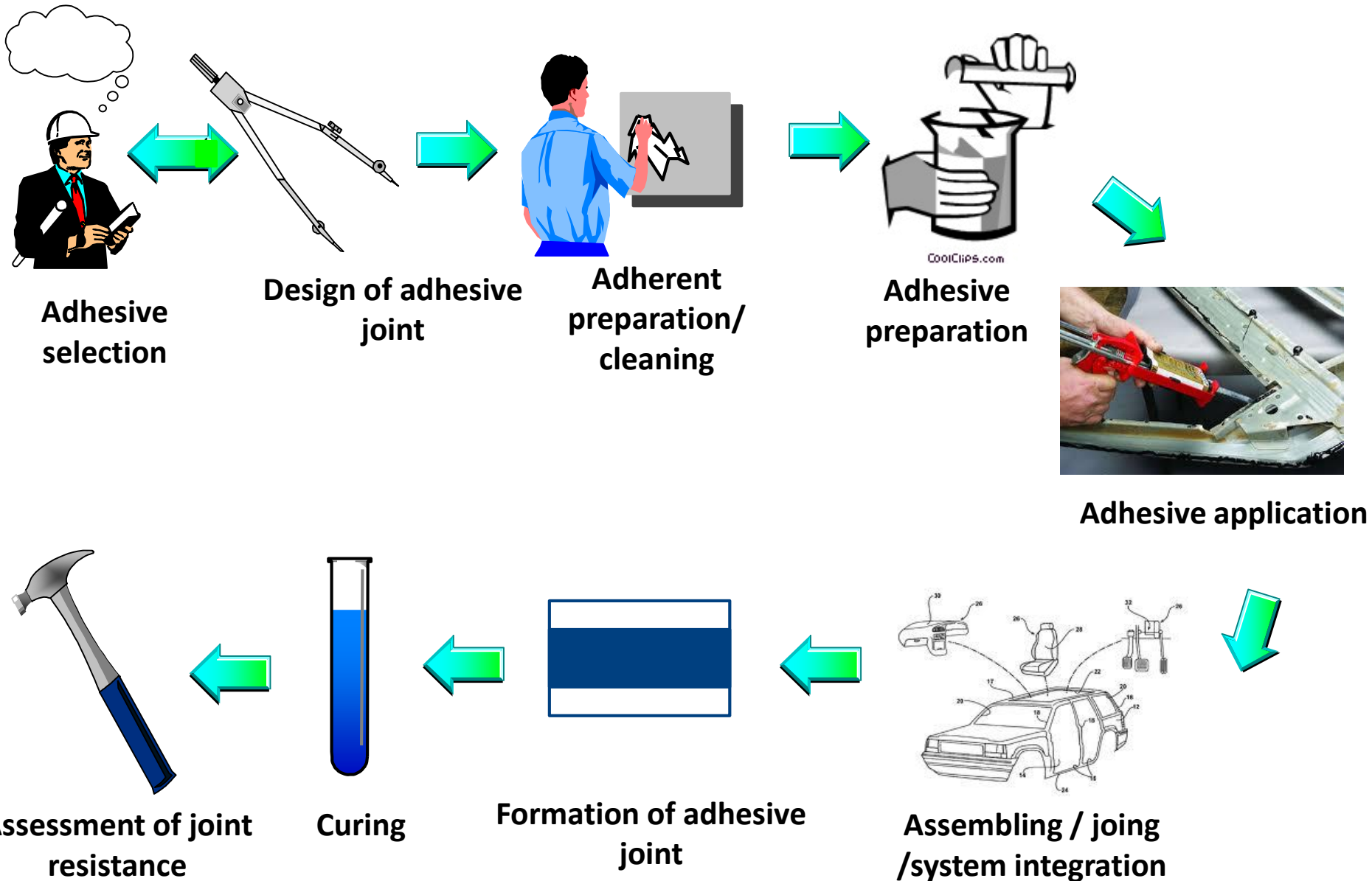
Epoxies

- Based on an **epoxide resin** plus **hardener** (**two component** system)
- Single component systems cured by heat usually at 180°C
- *Structural adhesives ✓*
- *high strength ✓*
- *excellent high temperature performance ✓*
- *Viscous and unsuitable for intricate work ✗*

Polyurathenes

- Usually **two component** system
- Very reactive with other polymers - good adhesion
- *Formulated for a wide range of applications*
- *For low temperature applications (< 65°C)*
- *Flexible and tough*
- *Sensitive to moisture and chemical environments*

4. Design and processing and considerations



4. Design and processing and considerations

Formation of an Adhesive Bond

1. Adhesive selection
2. Design of adhesive joint
3. Adherent preparation
 - Control of roughness, cleanliness and contamination
 - Stabilizing surface layers (pretreatment)
4. Adhesive preparation
 - Storage, mixing ect.
5. Adhesive application
 - Brushing flowing, film lay-up, ect
6. Joint assembly
 - Jigging, application of pressure, etc
7. Formation of adhesive joint
8. Curing
9. Testing

4. Design and processing and considerations



Adhesive selection

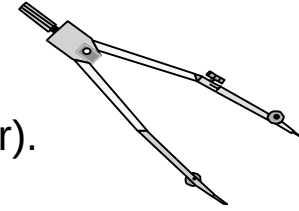
1. Adhesive selection

- Inherent ability for adhesion to the substrate
- Stresses experienced in service conditions
- Environment (temperature, chemical environment)
- Compatible coefficients of thermal expansion
- Special requirements (Conductivity, Appearance)
- Production restrictions
- Cost
- Health and Safety Factors (toxicity)

4. Design and processing and considerations

2. Design of Adhesive Joint

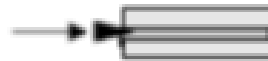
- Studying the stress distribution (shape, size, thickness of the pieces together). Adapt design of joint to the advantages of the adhesive.
 - Adhesives should ideally be loaded in **shear** or **compressed peel**
 - The design of the joint should **minimize tensile peel and tensile loading**
- Large bonding surfaces between the members to be joined are desirable



NO



Tension
(Butt Joints in Thin Sheet)

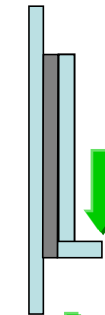
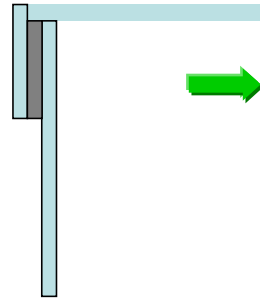
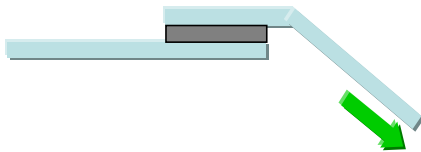


Cleavage

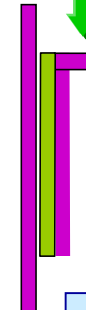
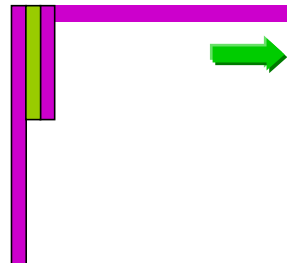
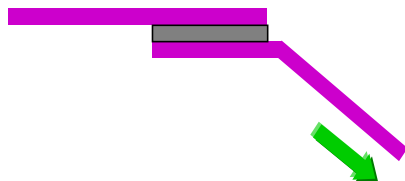


Peel

YES



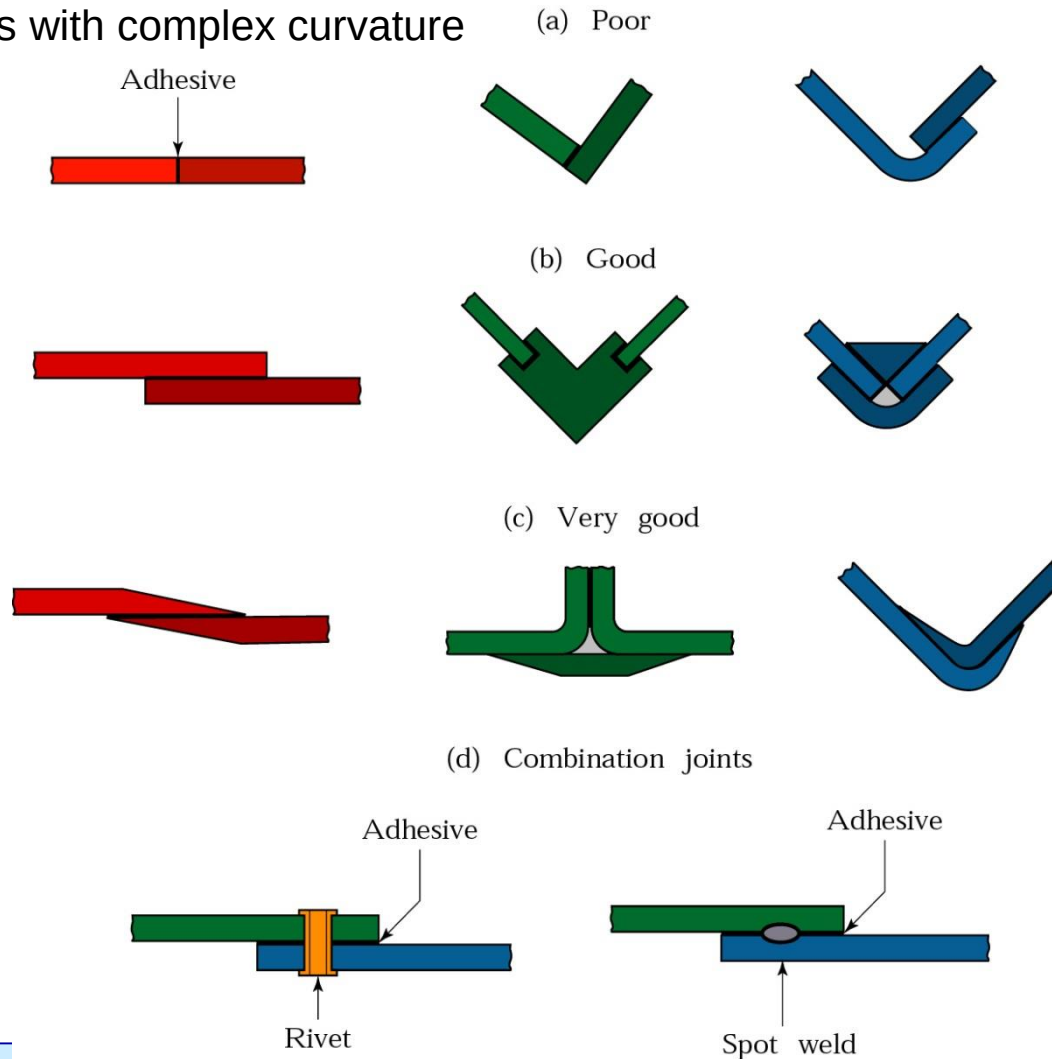
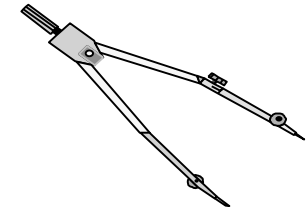
NO



4. Design and processing and considerations

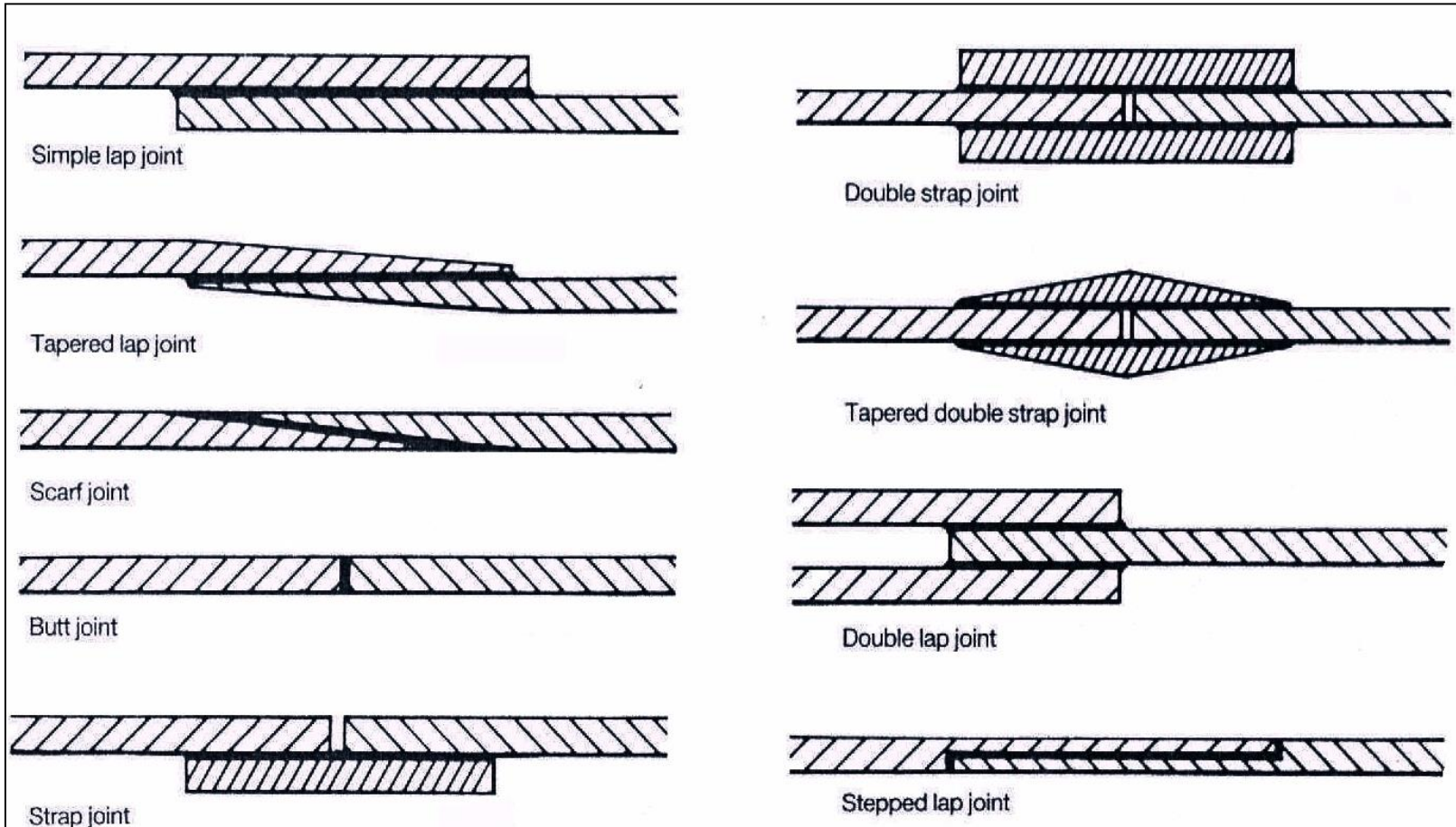
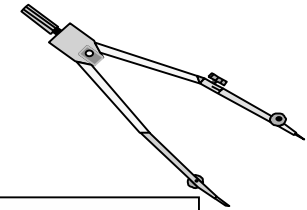
2. Design of Adhesive Joint

- Subassemblies designed to minimize adhesive operations.
- Avoid parts with complex curvature



4. Design and processing and considerations

2. Design of Adhesive Joint



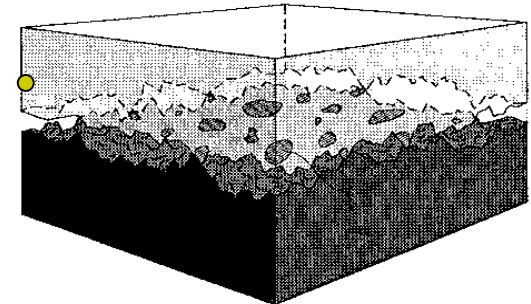
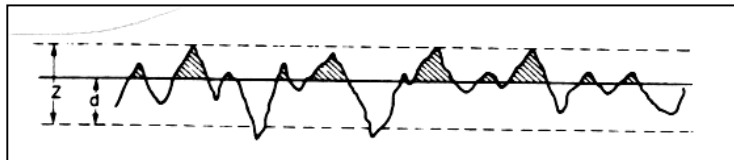
4. Design and processing and considerations

3. Adherent Preparation

- **Surface roughness** : appropriate for the substrate
- **Wettability**: the fluid has to fill the surface roughness. Must have lower surface tension than the substrate
- **The viscosity** of the adhesive must be adequate to enable the wetting of the substrate roughness



The adhesive has to adapt to the surface and then it has to harden



4. Design and processing and considerations

3. Adherent Preparation

Objectives

- ◆ adhesive substrate interface
- ◆ Facilitate reproducibility of results
- ◆ Avoid formation of low adhesion films (oxides)

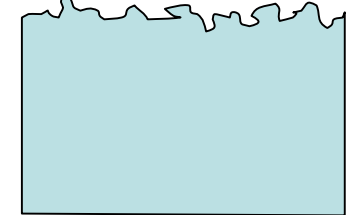
Surface Treatment:

- ◆ Removal of surface contaminants
 - **Mechanical methods** (abrasion). Oxide removal
 - **Chemical methods** (solvents, acetones, alcohols, etc): degreasing
- ◆ Physical or chemical modification of surface:
 - Appropriate **surface roughness**
 - Improve adherence: **primers**

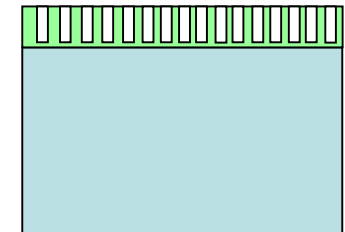


Surface Roughness

Grit blast



Anodized



Surface Cleanliness

4. Design and processing and considerations

3. Adherent Preparation

Good surface contact → creation adhesion → **PRIMERS**



Primers: a low viscosity adhesive that is applied before making the adhesive joint in order to improve adherence

- prevent capillary absorption of adhesive solution in porous adherent
- optimize interaction between substrate and adhesive: increase wetting, reactivity

Objectives:

- Protect substrate surface
- optimize interaction between substrate and adhesive: increase wetting, reactivity
- Increase strength of union
- Improve stress transfer
- Facilitates processing flexibility
- Protect against UV radiation

4. Design and processing and considerations

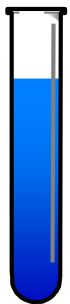
5. Adherent application



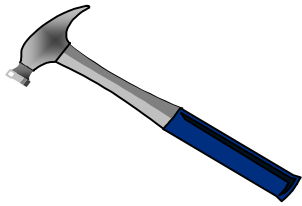
8. Curing

Curing in adhesives can be induced by:

- Anaerobic reaction: Reaction in the absence of O. Curing begins when trapped in a closed system in the presence of metal ions,
- Radiation or light (UV): photoinitiation
- Anionic reaction (cyanoacrylates): almost all substrates have absorbed water that contains OH which initiates polymerization by anionic reaction
- Activators (modified acrylics)
- Thermal treatment (epoxies)
- Biological components



5. Testing methods



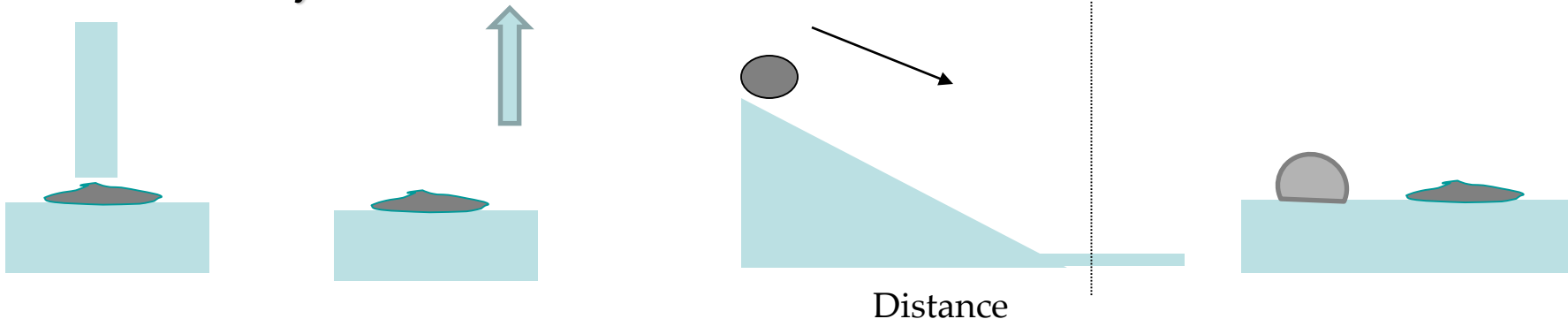
Assessment of joint resistance

The geometric factors that are considered to characterize an adhesive:

- the width of the adhesion
- the overlapping length
- thickness of adhesive and substrate.

1. TACK: Instantaneous Adhesion

- Ability to stick
- wettability

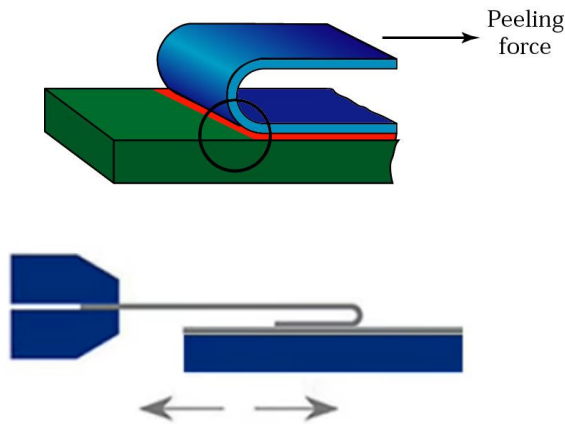


5. Testing methods

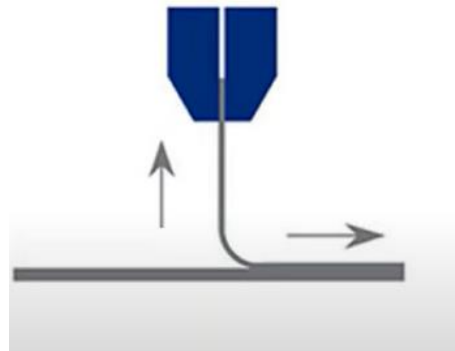
2. Adhesive Peeling Test: gives compatibility of the tape with a surface.

Depends: substrate properties (roughness, nature), peeling angle, rate P on the tape, d type adhesive mechanism, T .

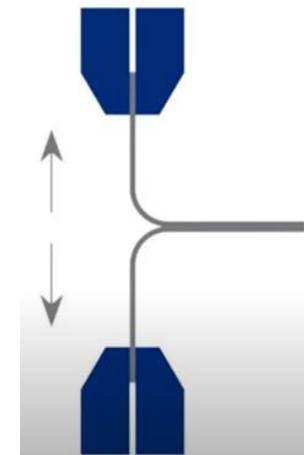
This test is similar to peeling adhesive tape from a solid surface. Adhesive joints should not be subjected to this type of loading in service.



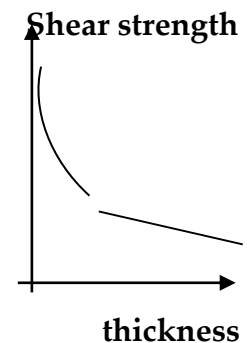
180° Peel Test



90° Peel Test

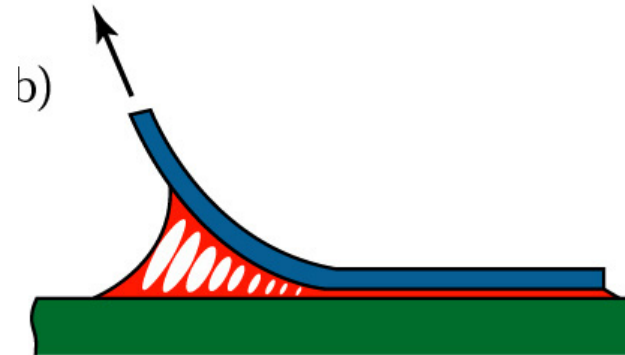
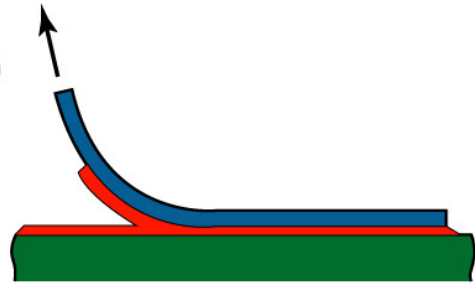


T-Peel Test

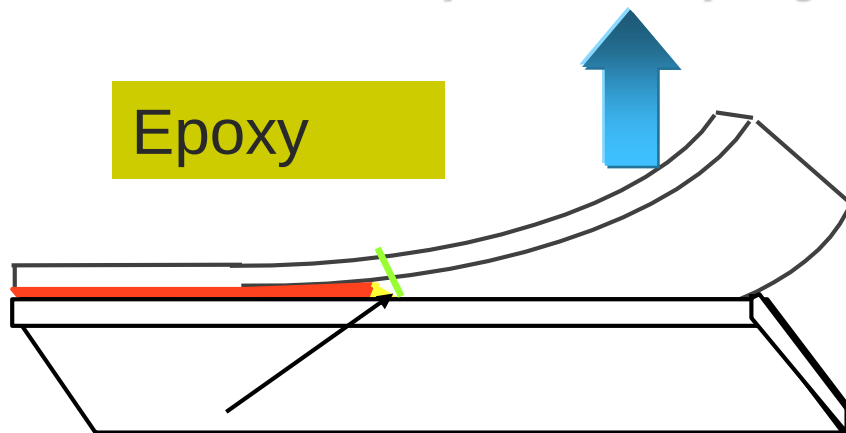


5. Testing methods

Effect of peeling on an adhesive joint

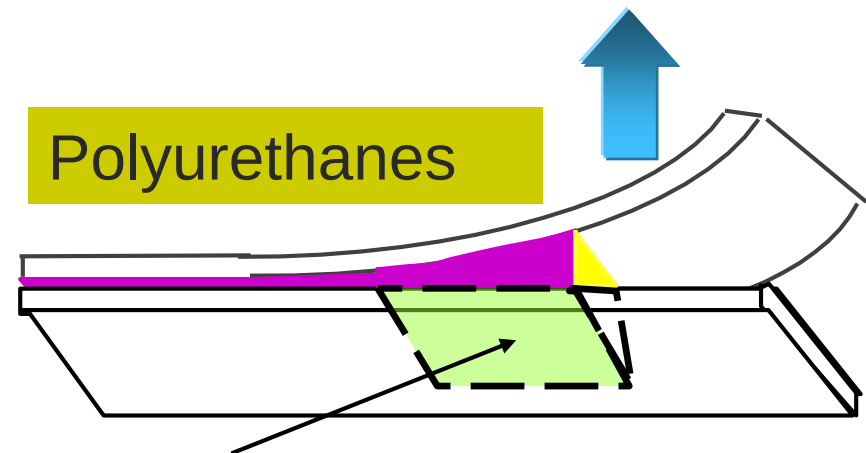


Characteristic behavior of **a) brittle** and **b) tough** adhesives in a peeling test.



Epoxy

Adhesive line standing stresses
RIGID: Only acts the line in
 adhesion front

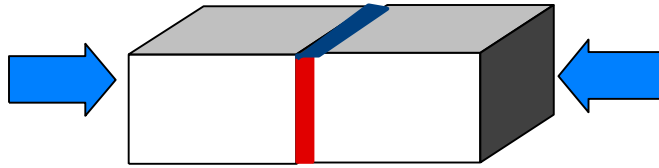


Polyurethanes

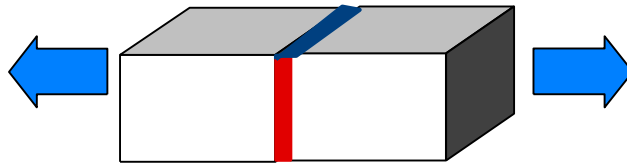
Volume of **FLEXIBLE** adhesive
 withstanding stresses
More surface supports the stress

5. Testing methods

Normal stresses



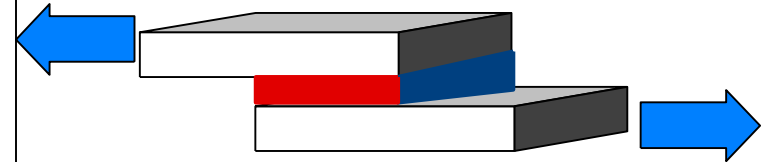
compression



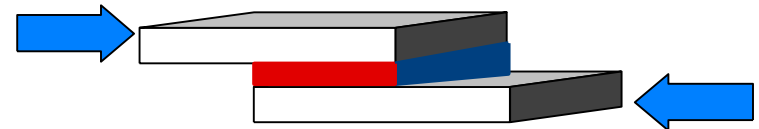
tension

Stresses are uniformly distributed in the joint. Structural adhesives must have high strength

Shear stresses



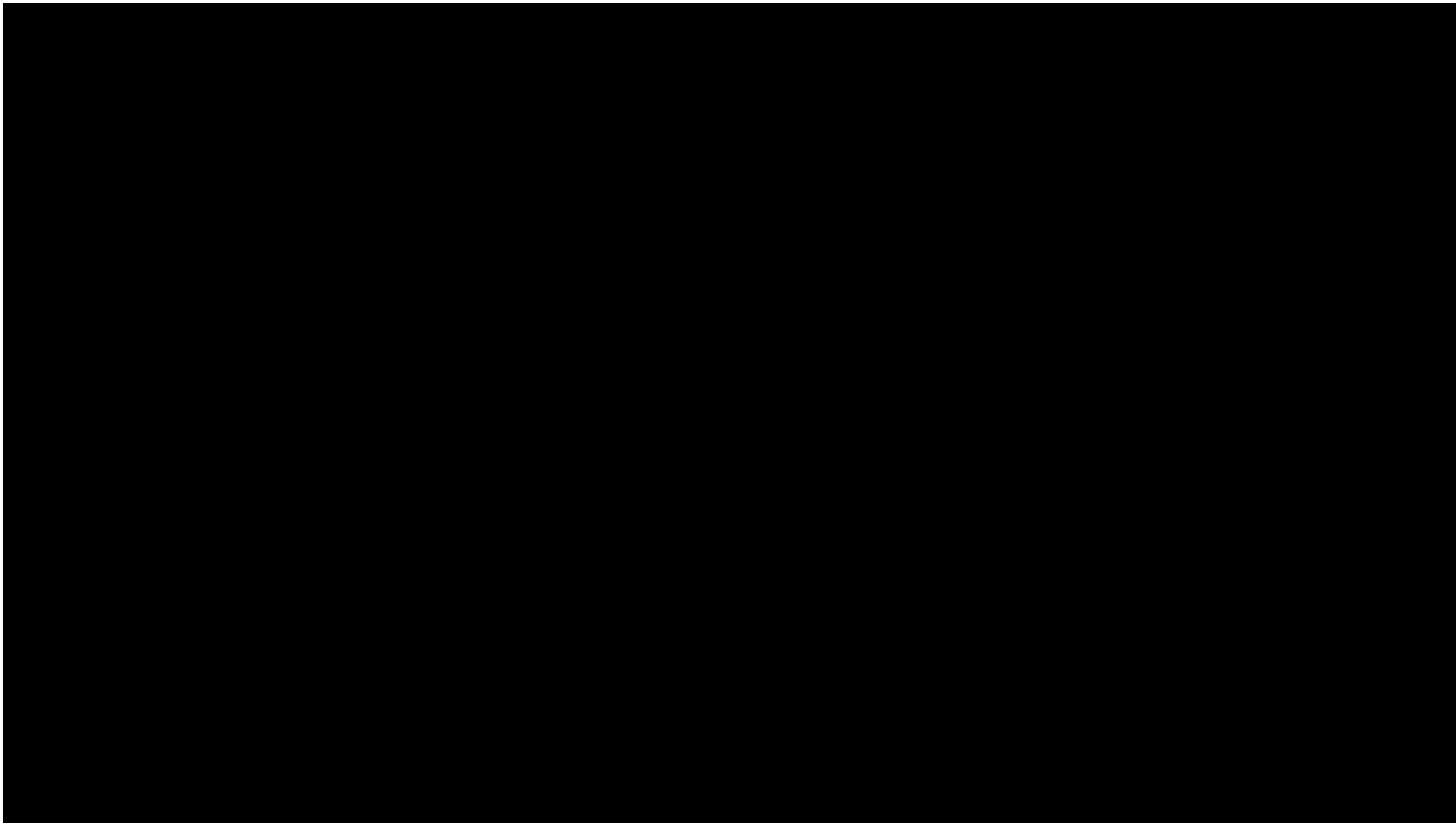
shear



Adhesive joint operates on the whole surface

5. Testing methods

<https://youtu.be/PPXVAJxe3f4>



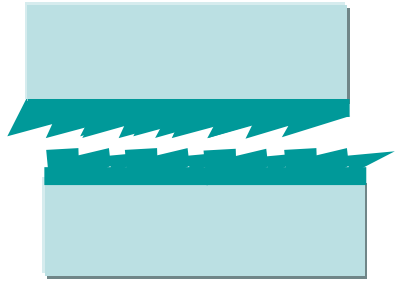
<https://www.youtube.com/watch?v=PFWN0rQmtp4>

<https://www.youtube.com/watch?v=I1SWCqNnE7c>

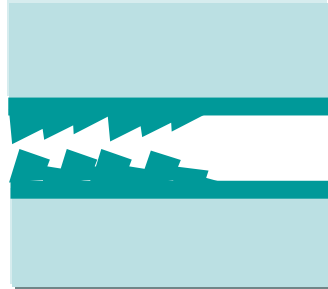
6. Modes of failure: Adhesive vs. Cohesive

Modes of failure

- Adhesive failure: failure that occurs between two phases
- Cohesive failure: failure that occurs within a phase



Cohesive failure



50% Cohesive failure

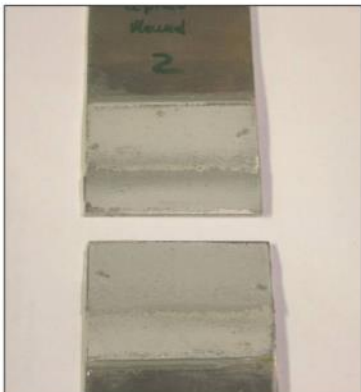


Adhesive failure



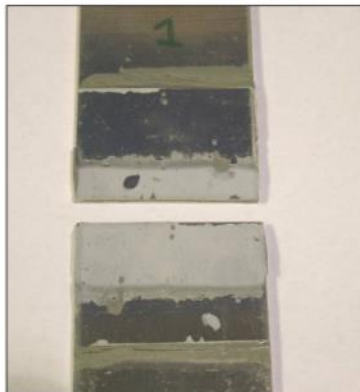
Substrate failure
(Over-dimension of adhesive)

Rotura Cohesiva



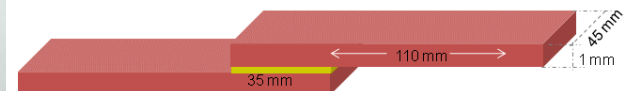
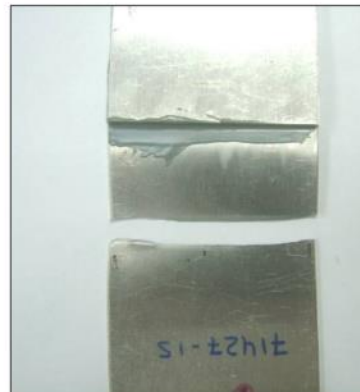
Adhesivo - Adhesivo

Rotura Adhesiva



Sustrato - Adhesivo

Rotura del Material



Tensile shear tests (**UNE-EN-1465**)

6. Modes of failure: Adhesive vs. Cohesive

Acceptable Failure Modes



Cohesive Failure
of Adhesive



Surface Ply Delaminations
(Composite Adherends)



Adherend Failure
(Static or Fatigue)

Unacceptable Failure Modes



Adhesion Failure of Adhesive/
Adherend Interface



Adhesive Peel
(Excessive Out-of-Plane Loads)



Adhesive Creep
(Flow)

7. Degradation mechanisms

Adhesive joints degrade gradually and fail

Main problem of adhesive bond is its **degradation**, that reduces properties and provokes failure

FACTORS affecting bond DURABILITY

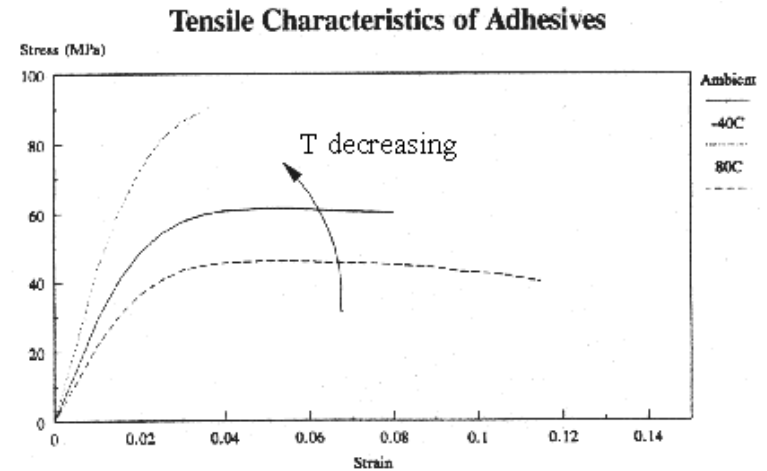
1. **Temperature:** Modify interface properties. $\uparrow T \Rightarrow \downarrow$ adherence. Differential thermal expansions $\Rightarrow$ residual stresses
2. **Moisture:**
 - a) Adhesive gets more plastic
 - b) Hydrolysis
 - c) Micro-cracking.

$\Rightarrow$ Use of fillers (mineral fillers, sealants, etc...)
3. **Electrochemical effects** (corrosion of adherents)
4. Combination of **Temp + moisture + mechanical loads**
5. **Aging:** Due to UV light and T. (i) If O_2 diffuses inside the adhesive: [polymer-O-O-H] (ii) with T $\rightarrow$
 $R-O\cdot + \cdot OH = \text{Free radical}$ destroying polymer chains

7. Degradation mechanisms

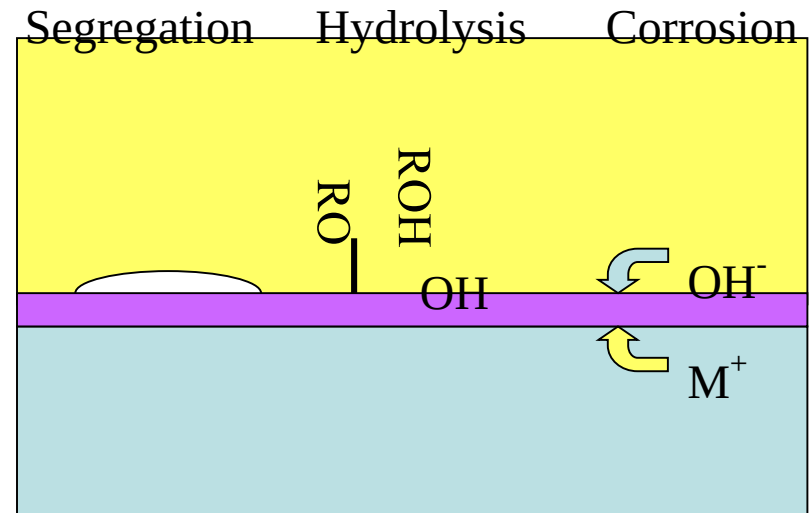
Changes in temperature modify interface properties:

- $T \uparrow, \downarrow$ adherence
- Differences in the coefficient of thermal expansion cause σ_{residual}



Effect of Moisture on the Joint

- Weakens adhesive material (becomes more plastic)
- Displaces the adhesive bond at the interface by preferential attraction of water (segregation)
- Causes chemical change of adherents (e.g. by corrosion of metal or hydrolysis of oxide layer)
- Can cause microcracking
- Effect increase with temperature and humidity



Moisture Attack at the Interface

7. Degradation mechanisms

- Performance criteria
 - Mechanical criteria
 - Resistance to static loading
 - Resistance to fatigue
 - Resistance to impact

Physical-chemical criteria

- Temperature resistance
- Aging with temperature
- Resistance to moisture
- Resistance to chemical agents

8. Examples

- Aircraft components
 - Structural joints (epoxies)
- Automotive components
 - Trim, stiffeners (epoxies, acrylics)
- Railway carriages
 - Joining composites to steel
- Loudspeaker cones
 - Quick assembly (cyanoacrylates)
- Foot wear
 - Hot melt adhesives
- Shafts/bearings/studs
 - anaerobics



General Properties of Adhesives

Type	Comments	Applications
Acrylic	Thermoplastic; quick setting; tough bond at room temperature; two component; good solvent chemical and impact resistance; short work life; odorous; ventilation required	Fiberglass and steel sandwich bonds, tennis racquets, metal parts, plastics.
Anaerobic	Thermoset; easy to use; slow curing; bonds at room temperature; curing occurs in absence of air, will not cure where air contacts adherents; one component; not good on permeable surfaces	Close fitting machine parts such as shafts and pulleys, nuts and bolts, bushings and pins.
Epoxy	Thermoset; one or two component; tough bond; strongest of engineering adhesives; high tensile and low peel strengths; resists moisture and high temperature; difficult to use	Metal, ceramic and rigid plastic parts.
Cyanoacrylate	Thermoplastic; quick setting; tough bond at room temperature; easy to use; colorless.	“Crazy glue.”™
Hot melt	Thermoplastic; quick setting; rigid or flexible bonds; easy to apply; brittle at low temperatures; based on ethylene vinyl acetate, polyolefins, polyamides and polyesters	Bonds most materials. Packaging, book binding, metal can joints.
Pressure sensitive	Thermoplastic; variable strength bonds. Primer anchors adhesive to roll tape backing material, a release agent on the back of web permits unwinding. Made of polyacrylate esters and various natural and synthetic rubber	Tapes, labels, stickers.

General Properties of Adhesives (cont.)

(continued)

Type	Comments	Applications
Phenolic	Thermoset; oven cured, strong bond; High tensile and low impact strength; brittle, easy to use; cures by solvent evaporation.	Acoustical padding, brake lining and clutch pads, abrasive grain bonding, honeycomb structures.
Silicone	Thermoset; slow curing, flexible; bonds at room temperature; high impact and peel strength; rubber like	Gaskets, sealants.
Formaldehyde: -urea -melamine -phenol -resorcinol	Thermoset; strong with wood bonds; urea is inexpensive, available as powder or liquid and requires a catalyst; melamine is more expensive, cures with heat, bond is waterproof; resorcinol forms waterproof bond at room temperature. Types can be combined	Wood joints, plywood, bonding.
Urethane	Thermoset; bonds at room temperature or oven cure; good gap filling qualities	Fiberglass body parts, rubber, fabric.
Water-base -animal -vegetable -rubbers	Inexpensive, nontoxic, nonflammable.	Wood, paper, fabric, leather, dry seal envelopes.

Typical Properties and Characteristics of Chemically Reactive Structural Adhesives

TABLE 30.3

	Epoxy	Polyurethane	Modified acrylic	Cyanoacrylate	Anaerobic
Impact resistance	Poor	Excellent	Good	Poor	Fair
Tension-shear strength, MPa (10^3 psi)	15.4 (2.2)	15.4 (2.2)	25.9 (3.7)	18.9 (2.7)	17.5 (2.5)
Peel strength, N/m (lbf/in.)	< 525 (3)	14,000 (80)	5250 (30)	< 525 (3)	1750 (10)
Substrates bonded	Most materials	Most smooth, nonporous	Most smooth, nonporous	Most nonporous metals or plastics	Metals, glass, thermosets
Service temperature range, °C (°F)	−55 to 120 (−70 to 250)	−160 to 80 (−250 to 175)	70 to 120 (−100 to 250)	−55 to 80 (−70 to 175)	−55 to 150 (−70 to 300)
Heat cure or mixing required	Yes	Yes	No	No	No
Solvent resistance	Excellent	Good	Good	Good	Excellent
Moisture resistance	Excellent	Fair	Good	Poor	Good
Gap limitation, mm (in.)	None	None	0.75 (0.03)	0.25 (0.01)	0.60 (0.025)
Odor	Mild	Mild	Strong	Moderate	Mild
Toxicity	Moderate	Moderate	Moderate	Low	Low
Flammability	Low	Low	High	Low	Low

Source: Advanced Materials & Processes, July 1990, ASM International.